Prompt Engineering: Thinking Clearly With Claude

Learning Objectives

By the end of this session, you should be able to:

- Define prompt engineering as clarity of thinking and recall the three elements of strong prompts (problem, expectations, reasoning).
- Explain the Claude model family (Haiku, Sonnet, Opus) and the speed, cost, quality tradeoffs.
- Select an appropriate model tier for a given business task using model-to-task mapping.
- Differentiate Extended Thinking vs Adaptive Thinking and determine when each mode is suitable.
- Construct effective system prompts by specifying role, tone, constraints, and output format.
- Use Chain-of-Thought prompting to structure multi-step reasoning for complex decisions.
- Iterate and assess prompts using informal evaluation criteria: accuracy, clarity, usefulness, relevance.

Agenda

In this session, we'll discuss:

- Prompting as Clarity of Thought
- Problem Definition, Expectation Setting, and Reasoning Direction
- Claude Model Family Overview: Haiku, Sonnet, and Opus
- Model Selection Criteria (Speed vs. Cost vs. Quality)
- Extended and Adaptive Thinking Approaches
- Enterprise System Prompt Design (Role, Tone, Constraints, and Format)
- Chain-of-Thought Prompting for Multistep Decision Processes
- Prompt Iteration and Evaluation Criteria (Accuracy, Clarity, Usefulness, and Relevance)
- Multimodal Inputs and Connectors Ecosystem (Chrome, Excel, PowerPoint, and Slack)

Prompting Is About Clarity of Thinking

Claude is only as good as the clarity of your thinking

Defining the Problem | Setting Expectations | Guiding Reasoning

✗ Poor Prompt

"Analyze this"

Vague, no context, no expectations

✓ Better Prompt

"Analyze this and identify Top 3 risks for a CFO"

Clear audience, clear output, clear scope

Defining the Problem | **Setting Expectations** | **Guiding Reasoning**

The Claude Model Family

Haiku | Sonnet | Opus

Three Model Tiers

Haiku

- Speed and Scale
- Cheapest and Fastest
- Base Quality
- Best for high-volume, simple tasks

Sonnet

- Balanced
- Slightly more expensive
- Better Quality
- Best for everyday business tasks

Opus

- Deep Thinking
- Most expensive and slowest
- Top Quality
- Best for complex reasoning and strategy

Matching Models to Tasks

Haiku Tasks

- Classification and Tagging
- Data Extraction
- High-Volume Simple Tasks
- Quick Lookups
- Routing And Filtering

Sonnet Tasks

- Business Writing
- Data Analysis
- Daily Workflows
- Email Drafting
- Report Summaries

Opus Tasks

- Strategy and Planning
- Deep Research
- Long-form Writing
- Complex Reasoning
- Multi-Step Analysis

Always consider the tradeoff: Speed vs. Cost vs. Quality

Extended vs. Adaptive Thinking

Extended Thinking

- You control the reasoning depth
- Explicitly set how much Claude should think
- Best when you know the problem needs deep analysis
- More predictable token usage

Adaptive Thinking

- Claude adjusts automatically
- Scales reasoning based on complexity
- Simpler for most use cases
- Claude decides how deep to go

Email = minimal thinking | Pricing Strategy = deep reasoning

The Role of System Prompts

System Prompts define:

- Role: Who Claude should be
- Tone: How Claude should communicate
- Constraints: What Claude should avoid

This transforms Claude from a generalist into a clearly-defined specialist

Task

Context

Constraints

Output Format

Clearly demarcated sections eliminate vagueness and force specificity

Chain-of-Thought Prompting

Explicitly request step-by-step reasoning for complex problems

Example: "Should We Increase SaaS Prices?"

Without CoT

"Should we increase prices?"

- Generic, shallow answer
- No structured reasoning

With CoT

Structured steps:

1. Analyze revenue
2. Evaluate benchmarks
3. Assess sensitivity
4. Identify risks
5. Recommend

CoT + Extended Thinking

- Deep reasoning on each step
- Each point is properly analyzed with effort
- Best for vaguely-scoped, reasoning-oriented prompts

Prompt Iteration and Evaluation

Expect ~50% of expected output on the first attempt. Requesting modifications is normal.

Text is information-sparse for conveying expectations—prompting is always iterative

Accuracy | **Clarity** | **Usefulness** | **Relevance**

Don't just accept outputs—informally evaluate every response against these criteria

Connectors Ecosystem

Chrome
Browse and interact with web content

Excel
Analyze and edit spreadsheet data

PowerPoint
Create and edit presentations

Slack
Access messages and channels

Why Connectors Matter:
- Real-time data access and cross-tool workflows
- Creates and edits actual files (not just text output)
- No more manual copy-paste integration

Key Takeaways

01

Prompt With Clarity

Define the problem, set expectations, and guide Claude's reasoning explicitly.

02

Match Model to Task

Haiku for speed, Sonnet for balance, Opus for depth. Always consider cost vs quality.

03

Connect Claude to Your Tools

Connectors let Claude work inside your everyday tools - not just in a chat window.

Summary

Here's a brief recap:

- Prompt engineering is primarily thinking clearly and expressing that thinking in a way the model can follow.
- Strong prompts improve outputs by defining the problem, setting expectations, and guiding reasoning.
- Claude tiers serve different needs: Haiku (fast/cheap), Sonnet (balanced), Opus (deep/complex reasoning). Choose based on speed, cost, and quality.
- Reasoning depth can be influenced via Extended Thinking (you control depth) or Adaptive Thinking (Claude adjusts automatically).
- System prompts turn Claude from a generalist into a specialist by enforcing role, tone, constraints, and output format.

Summary

Here's a brief recap:

- Chain-of-Thought prompting helps tackle complex decisions with a structured, step-by-step approach.
- Prompting is iterative; evaluate outputs using accuracy, clarity, usefulness, and relevance.
- Connectors enable Claude to work across tools and workflows (e.g., spreadsheets, slides, browser, Slack).

Learning Outcomes

You should now be able to:

- List the core components of an effective prompt (problem, expectations, reasoning).
- Describe Haiku/Sonnet/Opus and summarize their speed, cost, and quality tradeoffs.
- Map task types (classification, writing, strategy, multi-step analysis) to the best-fit Claude model tier.
- Choose between Extended vs Adaptive Thinking based on task complexity and predictability needs.
- Write system prompts that specify role, tone, constraints, and output format for enterprise contexts.
- Structure complex problem-solving prompts using Chain-of-Thought to reach a justified recommendation.
- Improve prompts through iteration and judge responses against accuracy, clarity, usefulness, and relevance.